

Homework 8

Deadline: April 28th, 2022

Self-test 7.3

Self-test 5.3 Suppose that 2.0 mol H_2 at 2.0 atm and 25°C and 4.0 mol N_2 at 3.0 atm and 25°C are mixed at constant volume. Calculate $\Delta_{\text{mix}}G$. What would be the value of $\Delta_{\text{mix}}G$ had the pressures been identical initially? $[-9.7 \text{ kJ}, -9.5 \text{ kJ}]$

7.4a

5.1(a) The partial molar volumes of acetone (propanone) and chloroform (trichloromethane) in a mixture in which the mole fraction of CHCl_3 is 0.4693 are $74.166 \text{ cm}^3 \text{ mol}^{-1}$ and $80.235 \text{ cm}^3 \text{ mol}^{-1}$, respectively. What is the volume of a solution of mass 1.000 kg?

7.5a

5.2(a) At 25°C , the density of a 50 per cent by mass ethanol–water solution is 0.914 g cm^{-3} . Given that the partial molar volume of water in the solution is $17.4 \text{ cm}^3 \text{ mol}^{-1}$, calculate the partial molar volume of the ethanol.

7.12 a

5.8(a) Consider a container of volume 5.0 dm^3 that is divided into two compartments of equal size. In the left compartment there is nitrogen at 1.0 atm and 25°C ; in the right compartment there is hydrogen at the same temperature and pressure. Calculate the entropy and Gibbs energy of mixing when the partition is removed. Assume that the gases are perfect.

5.5(b) The vapour pressure of 2-propanol is 50.00 kPa at 338.8°C , but it fell to 49.62 kPa when 8.69 g of an involatile organic compound was dissolved in 250 g of 2-propanol. Calculate the molar mass of the compound.

7.10 a

5.6(a) The addition of 100 g of a compound to 750 g of CCl_4 lowered the freezing point of the solvent by 10.5 K. Calculate the molar mass of the compound.

7.15a

5.11(a) Use Henry's law and the data in Table 5.1 to calculate the solubility (as a molality) of CO_2 in water at 25°C when its partial pressure is (a) 0.10 atm, (b) 1.00 atm.

7.20a

5.15(b) Given that $p^*(\text{H}_2\text{O}) = 0.02308 \text{ atm}$ and $p(\text{H}_2\text{O}) = 0.02239 \text{ atm}$ in a solution in which 0.122 kg of a non-volatile solute ($M = 241 \text{ g mol}^{-1}$) is dissolved in 0.920 kg water at 293 K, calculate the activity and activity coefficient of water in the solution.